# Serin Lee, Ph.D.

Assistant Professor, Department of Industrial & Management Engineering
Incheon National University, Incheon, Republic of Korea

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## EDUCATION

**University of Washington (UW)**, Seattle, USA

Ph.D. in Industrial Engineering — Jun. 2024
Thesis: "Modeling Healthcare Policy: From Calibration to Optimization"

M.S. in Industrial Engineering — Jun. 2021

**Seoul National University (SNU)**, Seoul, Republic of Korea

B.S. in Industrial Engineering — Aug. 2018

## RESEARCH INTERESTS

Healthcare, Modeling & Simulation, Simulation Optimization, Digital Twin, Agent-based Model, Pharmacometrics, Model Calibration, Disease Modeling

## EMPLOYMENT HISTORY

**Assistant Professor**, Incheon National University (INU) — Sep. 2025 – Present
Department of Industrial and Management Engineering

**Postdoctoral Research Fellow**, Fred Hutchinson Cancer Center — Nov. 2024 – Jul. 2025
Biostatistics, Bioinformatics, and Epidemiology Program

**Research Scientist**, Bill & Melinda Gates Foundation — Jun. 2023 – Feb. 2024
Institute for Disease Modeling, Global Health division

**Researcher**, Stanford University — Jun. 2021 – Sep. 2021
Department of Medicine, Center for Primary Care and Outcomes Research

**Undergraduate Researcher**, Seoul National University — Jun. 2017 – Dec. 2018
Center for Convergence Approaches in Drug Development

**Summer Intern**, Pfizer Pharmaceuticals Korea — Jun. 2016 – Aug. 2016
Preventive Health Management Division

## HONORS AND AWARDS

Industrial & Systems Engineering Faculty Endowed Fellowship, UW — 2021 – 2022
Lee B. Lusted Prize Finalist, Society for Medical Decision Making — 2020 & 2021
Graduate School Fund for Excellence and Innovation Top Scholar Award, UW — 2019 – 2020
Merit-based scholarship, SNU — 2017
National Scholarship For Science and Engineering, Korea Student Aid Foundation — 2016

## PUBLICATIONS

### Work in Progress

1. **Lee S**, Dimitrov D, Anderson P, Moore M. Dried blood spots measurements as indicator of HIV PrEP adherence and protection: Insights from pharmacokinetic modeling. Accepted at AIDS and Behavior
2. **Lee S**, Maneekul P, Zabinsky ZB, Liu S. A tutorial on explainable post-calibration analysis for interpreting policy outcomes. Under review

### Journal Papers

1. Moore M, **Lee S**, Matrajt L, Stansfield SE, Tao L, Yang J, Mezzio D, Jarrett J, Castro JC, Hsiao A, Zachry W, Dimitrov D. Modelling the 5-Year Impact of Twice-Yearly Injectable Lenacapavir for HIV Pre-Exposure Prophylaxis in the United States. Journal of the International AIDS Society. 2026;29(8):e70188
2. Weyant C, **Lee S**, Goldhaber-Fiebert JD. Reinforcement Learning-Based Control of Epidemics on Networks of Communities and Correctional Facilities. Medical Decision Making. 2026;46(2):216-25
3. **Lee S**, Zabinsky ZB, Liu S. Optimizing vaccination campaign strategies considering societal characteristics. Health Care Management Science. 2025;28(1):84-98
4. Stuart RM, Cohen JA, Kerr CC, Mathur P, National Disease Modelling Consortium of India, Abeysuriya RG, Zimmermann M, Rao DW, Boudreau MC, **Lee S**, Yang L, Klein DJ. HPVsim: An agent-based model of HPV transmission and cervical cancer. PLOS Computational Biology. 2024 07;20(7):1-17
5. **Lee S**, Zabinsky ZB, Wasserheit JN, Ross JM, Chen S, Liu S. Impact of vaccination and nonpharmaceutical interventions with possible COVID-19 viral evolutions using an agent-based simulation. AJPM Focus. 2024;3(1):100155
6. Weyant C, **Lee S**, Andrews JR, Alarid-Escudero F, Goldhaber-Fiebert JD. Dynamics of respiratory infectious diseases in incarcerated and free-living populations: A simulation modeling study. Medical Decision Making. 2022
7. **Lee S**, Zabinsky ZB, Wasserheit JN, Kofsky SM, Liu S. COVID-19 pandemic response simulation in a large city: impact of nonpharmaceutical interventions on reopening society. Medical Decision Making. 2021;41(4):419-29

### Peer-Reviewed Conference Proceedings

1. **Lee S**, Maneekul P, Zabinsky ZB. Representative calibration using black-box optimization and clustering. In: 2023 Winter Simulation Conference (WSC); 2023. p. 3669-80

## CONFERENCE PRESENTATIONS

### Oral Presentation

1. **Lee S**, Liu S, Zabinsky ZB. Optimizing Vaccination Campaign Considering Societal Characteristics. Daejeon, Republic of Korea; 2025

2. **Lee S**, Liu S, Zabinsky ZB. Optimizing Vaccination Campaign Strategies Considering Societal Characteristics. Seattle, WA; 2024
3. **Lee S**, Liu S, Zabinsky ZB. Optimal Vaccine Promotion Campaigns Considering Disease Transmission and Opinion Propagation with Fairness. Phoenix, AZ; 2023
4. **Lee S**, Liu S, Zabinsky ZB. Optimal Budget Allocation to Vaccine Promotion Campaigns Considering Disease Transmission and Opinion Propagation. Toronto, Canada; 2023
5. **Lee S**, Liu S, Zabinsky ZB. Advancing Simulation Methodology for Identifying Optimal Healthcare Policy during the COVID-19 Pandemic. Seattle, WA; 2023
6. **Lee S**, Liu S, Zabinsky ZB. Optimal Vaccination Campaigns under Opinion and Physical Networks. Indianapolis, IN; 2022
7. **Lee S**, Zabinsky ZB, Liu S. Impact of Vaccination and Relaxation of Non-Pharmaceutical Interventions on Controlling COVID-19 Using Agent-Based Simulation. Virtual; 2021. **Finalist for the Lee B. Lusted Award**
8. **Lee S**, Zabinsky ZB, Liu S. Using Agent-Based Simulation to Optimize Vaccination Schedules and Non-Pharmaceutical Interventions to Control COVID-19 Outbreaks. Anaheim, CA; 2021
9. **Lee S**, Zabinsky ZB, Liu S. Using Agent-Based Simulation to Analyze How to Achieve Herd Immunity to SARS-CoV-2 with Vaccination and Non-Pharmaceutical Interventions. Virtual; 2021
10. **Lee S**, Zabinsky ZB, Kofsky SM, Liu S. Impact of Population Health Interventions for a Measles Outbreak in a Low-Vaccination Area. Virtual; 2020
11. **Lee S**, Zabinsky ZB, Kofsky SM, Liu S. Agent-Based Simulation of COVID-19 Society Reopening Plans in an Urban Setting. Virtual; 2020
12. **Lee S**, Zabinsky ZB, Liu S. Agent-Based Simulation Study of Reopening Scenarios for COVID-19 in a Large Urban Area. Virtual; 2020. **Finalist for the Lee B. Lusted Award**

**Invited Seminar**

1. **Lee S**. Data-Driven Approaches to Support Clinical Trials. Ochang, Republic of Korea; 2026
2. **Lee S**. Evaluating DBS Snapshots as a Benchmark for HIV PrEP Adherence and Protection in HPTN 067. Seoul, Republic of Korea; 2026
3. **Lee S**. Introduction to Agent-Based Modelling. Seoul, Republic of Korea; 2020

**Poster Presentations**

1. **Lee S**, Liu S, Zabinsky ZB. Optimal Vaccination Campaigns under Disease Transmission and Opinion Propagation. Seattle, WA; 2022
2. **Lee S**, Zabinsky ZB, Liu S. Optimizing COVID-19 Vaccine Prioritization Policies with Non-Pharmaceutical Interventions Using Agent-Based Simulation. Virtual; 2021
3. **Lee S**, Jeon Y, Chung TK, Kim M, Kim S, Lee H. Dimensionality Reduction Model for Common Eligibility Criteria to Effectively Use Electronic Medical Records: A Pilot Study. Seoul, Republic of Korea; 2018

## TEACHING

**Undergraduate courses at INU**

| | |
|---|---|
| Simulation | 2026 Fall |
| Data Literacy | 2026 Fall |
| Database | 2025 Fall, 2026 Spring |
| Capstone Design | 2026 Spring |

**Graduate courses at INU**

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|---|---|
| AI Programming | 2026 Spring |
| Digital Twin | 2025 Fall |

## PROJECTS

Digital Twin-based Virtual Clinical Trial for Infectious Diseases:Effectiveness Prediction and Protocol Optimization

*PI*, National Research Foundation of Korea — Mar. 2026 – Feb. 2029

Agent-Based Simulation Research Using Publicly Available Data

*PI*, INU — Apr. 2026 – Apr. 2027

Studies on data-driven simulation and modeling

*PI*, INU — Oct. 2025 – Apr. 2027

## PROFESSIONAL ACTIVITIES

**Reviewer**

Medical Decision Making
Health Care Management Science
Journal of Medical Internet Research

**Member**

Institute for Operations Research and the Management Sciences (INFORMS)
Society for Medical Decision Making (SMDM)
Korean Institute of Industrial Engineers (KIIE)